

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Alpha presents <u>no</u> danger or it can't pass through the skin (1) Beta and gamma can get into the body (1) Beta is more <u>ionising</u> [than gamma so more dangerous] (1) Treat statements about cancer as being neutral	3			3		
	(b)	(i)		Finding the ratio of LLW to reactors in UK = $\frac{30\,100}{15} = 2007$ (1) For France, ratio = $\frac{87\,000}{58} = 1\,500$ so not valid (1) [Conclusion must be present to award 2 marks] Alternative: Ratio of reactors France to UK is $\frac{58}{15} = 3.87$ (1) Ratio of LLW France to UK = $\frac{87\,000}{30\,100} = 2.89$ so not valid (1) [Conclusion must be present to award 2 marks] Alternative: Ratio of reactors France to UK is $\frac{58}{15} = 3.87$ (1) Expected LLW in France: $30\,100 \times 3.87 = 116\,487$ so not valid (1) [Conclusion must be present to award 2 marks]			2	2	2	
		(ii)	I	The presence of volcanoes (1) Creates a danger of leakage from the containers (1)			2	2		
			II	Large population in a small area / population density (1) means there is less likelihood of remote areas being available as storage sites or being stored far away from people (1)			2	2		

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		(iii)		Low fuel <u>cost</u> [over time] (1) It now needs to include the cost of long term storage of the waste (which is very high) (accept reference to cost of decommissioning) (1)			2	2		
		(iv)		Either: Sent on a rocket into space (1) – adv-gone from Earth, disadv-costly/danger of take-off failure (1) Or: Very deep sea storage (Accept underwater) (1) – adv – well away from human activity, disadv – possible leakage in the long term (1) Or: Polar ice cap storage or remote island (1) – adv – well away from human activity – disadv – long term leakage etc (1) Or: Vitrification (1) – adv – easily stored, disadv – high cost (1)	2			2		
				Question 4 total	5	0	8	13	2	0